

SUFAL K.C.

kcsufal.com.np | linkedin.com/sufal-kc | github.com/sufal-kc | Balaju, Kathmandu

Summary

Electronics, Communication, and Information Engineering student focused on AI, computer vision and IoT based projects. Experience developing computer vision pipelines, predictive models, and data driven solutions using Python, OpenCV, and Scikit-learn. Eager to apply technical skills to develop innovative, real world solutions.

Technical Skills

- **Programming:** Python, C++, C, Java, JavaScript, SQL
- **Tools:** Git, GitHub, Linux, VS Code
- **AI/ML & Data:** Scikit-learn, OpenCV, MediaPipe, Pandas, NumPy, Matplotlib
- **Web Development:** HTML, CSS, React.js, Node.js, RESTful APIs
- **Embedded & IoT:** Arduino, ESP32, Wi-Fi CSI
- **Concepts:** Computer Vision, Machine Learning, IoT, Signal Processing, Object-Oriented Programming (OOP)

Projects

AI Powered Eye Controlled Assistive Mobility System | *OpenCV, MediaPipe, Python*

Hackathon Project

Jan 2026

- Developed a real time eye tracking and blinking detection system for hand free assistive mobility.
- Implemented movement detection logic and a fail safe kill switch to improve control reliability and user safety.
- Integrated joystick based manual control and ultrasonic obstacle detection to provide better navigation.

RF Wave CSI Based Motion Detection System | *Machine Learning, IoT, Wi-Fi CSI*

Research and Minor Project

Sep 2026

- Processed WiFi CSI amplitude and phase data from ESP32 for human position and motion detection.
- Applied signal processing and noise filtering to extract meaningful spatial features from raw CSI measurements.
- Developed near real time visualization of predicted human positions using a stickman figure interface.

Smart QR Code Attendance System | *Java, REST API, SQL*

Independent Project

May 2026

- Developed a full-stack attendance system supporting Admin and Student roles, with REST APIs for authentication and SQL based attendance management.
- Implemented secure RESTful APIs to handle user authentication and manage relational database (SQL) interactions efficiently.

Experience

Sagarmatha Electronics Engineering Society (College Robotics Club)

Vice President

Sanepa, Lalitpur

2024 – Present

- Led a Data Science learning initiative for 100+ learners, facilitating sessions on Python, data analysis, and AI/ML concepts.
- Organized technical workshops, ranging from a 7-days Arduino bootcamp to advanced computing sessions.
- Represented the institution at major tech events, robotics event, and national hackathons.

UNESCO MGIEP

Online Outreach Volunteer

Remote, India

Oct 2025 – Jan 2026

- Assisted in a research project, successfully acquiring 800 target participants within a 8 weeks deadline.
- Coordinated tasks and streamlined communication channels among a remote team of 8 research teammates.

Honors and Awards

- **Best Thematic Award** | Taranga Hardware Hackathon
- **1st Runner Up** | KathFest Hackathon 3.0
- **Top 60 Learner** | Leapfrog #60DaysOfLearning2026 Challenge

Volunteering

OrigamiforGood Nepal

Graphic Design Intern

2025

- Learner poster designing, origami pamphlet design and Animate poster using Canva

References

- **Er. Baikuntha K. Acharya** – Assistant Professor, Dept. of Computer and Electronics Engineering
Sagarmatha Engineering College, Kathmandu, Nepal